FRANCES (SU) LIU

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RESEARCH INTERESTS

Financial machine learning; financial econometrics and forecasting; high-frequency and alternative data; portfolio construction and risk modeling.

My research takes on the problems a quantitative investment team faces every day: finding signal in noisy, fast-moving data, deciding how to allocate capital, and measuring risk before it materializes. I approach them with machine learning and econometrics, informed by six years in industry.

EDUCATION

Doctor of Philosophy (PhD) in Finance *2024.02–2027.08 (expected)*
University of Technology Sydney (UTS), Sydney, Australia

Supervisors: Assoc. Prof. Vitali Alexeev (UTS, Finance), Assoc. Prof. Katja Ignatieva (UNSW, Actuarial) & Prof. Adam Clements (QUT, Finance)

Master of Science in Financial Engineering *2015.08–2017.06*
Stevens Institute of Technology, New Jersey, United States *GPA: 3.9/4.0*

Thesis: Systemic Risk Measurement: CoVaR and Copula Models (ARMA/TGARCH conditional moments, copula-based dependence across financial sectors, time-series CoVaR). Supervised by Assoc. Prof. Zhenyu Cui

Bachelor of Economics in Financial Engineering *2011.09–2015.06*
Harbin University of Commerce, Harbin, China

RESEARCH PAPERS

Reading Between the Trades: Neural Encoding Extended-Hours Tick Sequences for Volatility Forecasting
with Vitali Alexeev (UTS), Adam Clements (QUT), and Katja Ignatieva (UNSW) *working paper*

Reads each extended-hours session, usually reduced to one overnight return, as an irregularly spaced trade stream, using a permutation-invariant set encoder layered as a correction on a fixed HAR benchmark; tested first in simulation, with a model ladder separating what session realized variance captures from what the encoder adds.

Econometric Society Australasia Meeting (Adelaide, November 2026); 2nd Conference of the Australian and New Zealand Association of Econometricians (Sydney, November 2026)

When Does Sentiment Predict Volatility?
with Vitali Alexeev (UTS) and Katja Ignatieva (UNSW) *working paper*

Shows that the negligible average forecasting gain from media sentiment hides where it occurs: gains concentrate in low-persistence, high-volatility states and turn to losses elsewhere, a pattern a Monte Carlo study ties to predictive content that rises as persistence falls.

International Association for Applied Econometrics (IAAE) Annual Conference (Lisbon, June 2026); FIRN Annual Conference (Queensland, November 2026)

AWARDS AND SCHOLARSHIPS

UTS Business Doctoral Scholarship (full stipend) 2024–2027
UTS International Research Scholarship (full tuition) 2024–2027
Academic Excellence Award, Stevens Institute of Technology 2017

TEACHING EXPERIENCE

University of Technology Sydney *Sydney, Australia*
Teaching Assistant, Finance Discipline Group 2026–present

- 25881 *AI-integrated Sustainable Finance* (postgraduate)

Tutor, Finance Discipline Group 2024–present

- 25503 *Investment Analysis* (undergraduate)
- 25556 *The Financial System* (undergraduate)

INDUSTRY EXPERIENCE

Six years of applied quantitative research across systematic strategies, portfolio construction and asset allocation, risk modeling, structured-finance valuation, and research infrastructure, carrying models from hypothesis through backtesting to deployment.

Webull Financial LLC *Changsha, China*

Senior Quantitative Analyst 2020.04–2023.11

- Developed portfolio construction and asset allocation for a client-facing investment-advisory product spanning five risk profiles, combining screened ETF universes, mean-variance optimization, CAPM-implied returns, machine learning denoising, and covariance shrinkage
- Researched and implemented mid-frequency strategies across Chinese and US equity markets, including factor-based stock selection, sector allocation, ETF rotation, and equity-index strategies
- Took strategies from research and backtesting through paper trading to live deployment, with ongoing post-deployment performance monitoring, using S&P, FactSet, and Refinitiv data
- Designed and co-developed the shared research database and backtesting environment used by the 12-person private-fund quantitative research team

Pricing and Analytics (Shenzhen) Limited *Shenzhen, China*

Quantitative Analyst: Derivatives Pricing and Valuation 2017.12–2019.10

- Developed dynamic pricing, valuation, and risk models for property-backed REITs and CLOs issued by listed real-estate companies across successive issues; two models were registered as software copyrights
- Modeled underlying cash flows and calibrated prepayment assumptions from market-observable spreads, and applied Black-Scholes, risk-neutral and martingale pricing, extreme value theory, and Value at Risk for portfolio tail risk
- Implemented the pricing engines in C++ with Excel interfaces and VBA integration, with relative valuation against market comparables selected at city and listed-issuer level

COMPUTING AND LANGUAGES

Computing: Python (PyTorch, TensorFlow, Keras), R, SQL, C++; GPU/HPC cluster computing
Languages: English (fluent), Mandarin (native), Cantonese (fluent)